

3.2 Your First Program

The C++ 20/23/26 Masterclass

THE RUNNING EXAMPLE

Throughout this chapter we build **one small program**: a personal profile that greets you, asks a few questions, makes decisions, and - by the end - remembers your friends between runs.

- Nothing fancy on purpose
- One idea added per lecture, on top of the last
- The same excuse to meet every foundational building block

BEFORE WE TYPE ANYTHING

[SCREENSHOT]

Every lecture folder has its own `CMakeLists.txt`, already scaffolded.
Open the folder, don't create a new project by hand.

THE SMALLEST REAL PROGRAM



```
#include <print>

int main() {

    std::println("Hello there!");
    std::println("Welcome to your very first C++ program.");

    return 0;
}
```

3.2 / LINE BY LINE

WHAT'S ACTUALLY HAPPENING

LINE	WHAT IT DOES
<code>#include <print></code>	pulls in <code>std::println</code>
<code>int main()</code>	every C++ program starts here
<code>std::println(...)</code>	prints text, adds the newline for you
<code>return 0;</code>	tells the OS the program finished successfully

3.2 / RUN IT

BUILD AND RUN

Same three commands every lecture in this course:



```
cmake -B build -G Ninja .  
cmake --build build  
./build/rooster
```

3.2 / RUN IT

WHAT YOU SHOULD SEE

[SCREENSHOT]

TWO KINDS OF COMMENTS



```
// A one-line comment - the compiler ignores everything after //
```

```
/*
```

```
  A block comment.
```

```
  Can span multiple lines - handy for a longer explanation.
```

```
*/
```

Rule of thumb: a comment earns its place by explaining **why**, not by narrating what the code obviously already says.

Next: 3.3

Statements and Functions